Lithium-Magnesium Binary Alloy Anodes for Solid-State Batteries

Anne Jumper[1]

***Abstract* — Lithium-magnesium (Li-Mg) alloy anodes are a promising alternative to graphite and pure lithium anodes, offering a theoretical capacity of 3,350 mAh/g and improved mechanical and electrochemical stability. This paper provides an in-depth exploration of the properties of Li-Mg and how they affect its performance as an anode in solid-state batteries. Li-Mg forms a substitutional BCC solid solution in the Li-rich state, allowing for dual electron-ion conductivity in the anode and limiting dendrite formation. Increasing Mg content reduces grain size, increases hardness, and results in solid solution strengthening, while reducing Li diffusivity due to short-range order effects. The binary phase diagram constrains cycling to at most one phase transformation per half-cycle, restricting operation to the β-phase to eliminate transformation induced stresses. The primary failure mechanisms include void coalescence at the electrode-electrolyte interface during stripping and dendrite formation during plating, both of which are significantly reduced with the addition of Mg, making it a viable candidate for next-generation SSB anodes.**

I. Introduction

The development of advanced energy storage solutions is more critical than ever due to growing energy demands and economic and environmental calls for fossil-fuel independence. For current commercial applications, lithium-ion batteries (LIBs) are a common mode of energy storage for portable systems requiring a high energy density; most notably, they are crucial to the electric vehicle (EV) industry. However, LIBs are limited by slow charging times, short cycle life, limited driving range in EVs, and flammability of the liquid electrolyte. Lithium metal anodes are thought to have the potential to significantly enhance battery performance owing to their high theoretical capacity (3860 mAh/g). Implementation of a lithium metal anode in the LIB proves challenging due to its interactions with the liquid electrolyte; however, realization of the lithium metal anode is possible with a solid-state electrolyte (SSE) [1].

Despite the SSEs improved compatibility with lithium metal anodes, the widespread implementation of solid-state batteries (SSBs) with lithium metal anodes remains challenging due to lingering instability and dendrite formation issues. Dendrites are microscopic needle-like lithium structures which can penetrate the electrolyte and grow from one electrode to the other during cycling, resulting in internal short circuits and battery failure [1]. During charging, uneven lithium plating on the anode surface facilitates dendrite growth by creating regions with elevated local electronic conductivity, where solid lithium preferentially forms. Additionally, repeated lithium plating/stripping, volume expansion of the anode, and dendrite growth cause can create a thick, brittle solid-electrolyte interface (SEI). The SEI limits efficiency by consuming active material and severely limiting ion transport.

A proposed solution to mitigate dendrite growth and volume expansion associated with lithium metal anodes is by replacing them with a lithium-magnesium alloy. The properties of this alloy make it particularly effective. See Fig. 1, namely, Li-Mg alloys have higher electronic and ionic conductivity than Li metal alone. Magnesium is lithophilic, meaning lithium ions are drawn more evenly across the anode surface. The smoother microstructure of this lithium plating reduces preferential sites for dendrite growth. [2]

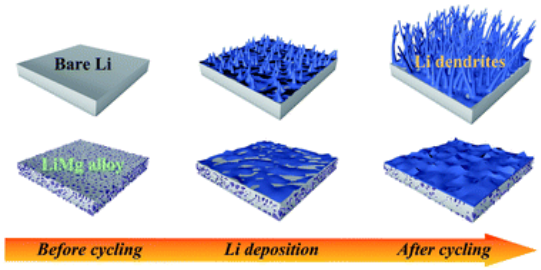


Fig 1. Difference in Li plating on Li vs. Li-Mg anodes after cycling [2]

This paper provides an in-depth exploration of the properties of lithium-magnesium alloy which make it an ideal anode candidate in commercial solid-state batteries.

II. Atomic Structure and Microstructure

Li-Mg forms a substitutional solid solution since both atoms have similar atomic radii (152 pm and 160 pm respectively), valences (+1 and +2 respectively), and both assume a BCC crystal structure in the lithium-rich state [3]. In the solid solution, Li and Mg atoms occupy random lattice sites in the BCC structure. Additionally, despite the difference in the electronegativity of the atoms (0.98 and 1.31), no intermetallic compounds form [4], and thus, there is no phase change stress or cracking, improving the life of Li-Mg anodes.

During lithiation and delithiation during cycling, the solid solution reaction dominates over the reconstitution reaction, i.e. Li generated at the anode surface diffuses inward rather than accumulating, which prevents dendrites [2]. The formation of a solid solution is preferable to a reconstitution reaction which requires high activation energy and results in significant change in the crystal structure, which can cause swelling or cracking of the anode during cycling, resulting in an unstable SEI layer and potentially mechanical failure [2]. A substitutional solid solution prevents energy waste and results in lower voltage hysteresis (voltage gap between charge and discharge) [2].

The Li-Mg alloy framework also acts as an electron and ion

[1]Anne Jumper is a student in the Mechanical Engineering Department at Tufts University, Medford, MA 02155 USA, (e-mail: anne.jumper@tufts.edu).

dual conductive host, meaning that both electrons and Li ions can move freely through the Li-Mg anode during cycling, promoting uniform current distribution across the anode surface which reduced preferential sites for lithium deposition and, thus, dendrite formation [2]. Not only does this encourage deeper Li intercalation during plating, but it also promotes better Li extraction during stripping, both of which prevent Li accumulation and dendrites [2]. Additionally, volume change is also inhibited by preventing ion accumulation. The increased exchange current density of Li-Mg (2.56 mA cm^{-2}) compared to pure Li (1.31 mA cm^{-2}) supports these facts [2]. This is indicative of faster charge transfer at the interface between electrode and electrolyte and thus, faster charge and discharge times.

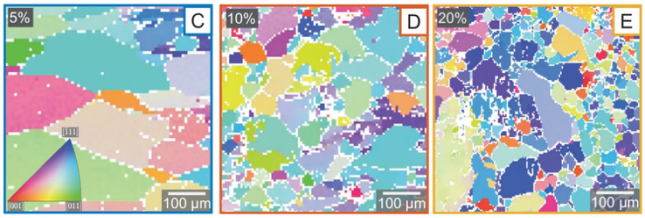


Fig 2. Representative EBSD maps of 5, 10 and 20 %at Mg alloys [5].

Looking at microstructural changes with increased Mg, electron backscattered diffraction mapping conducted by Aspinall et al. (shown in Fig. 2) of as-cast Li-Mg alloys cooled from 450ºC reveals grain size reduction with increasing Mg content. Mean grain diameters decrease from 116.7 ± 65.2 μm at 5 at% Mg to 50.6 ± 34.4 μm at 10 at% Mg to 33.5 ± 21.6 μm at 20 at% Mg. This grain refinement can be attributed to Mg solute partitioning, that is, magnesium physically blocks grain growth, forcing metal to form smaller and more refined crystals. However, slower cooling can cause complex, internal branch patterns to develop in the crystals [5]. For solid-state anodes, finer grains increase the density of grain boundaries, which act as fast diffusion pathways and vacancy sinks and may accelerate Li transport locally and absorb excess vacancies formed during stripping, preventing void nucleation at the electrode-electrolyte interface [5].

III. Mechanical Properties and Strengthening

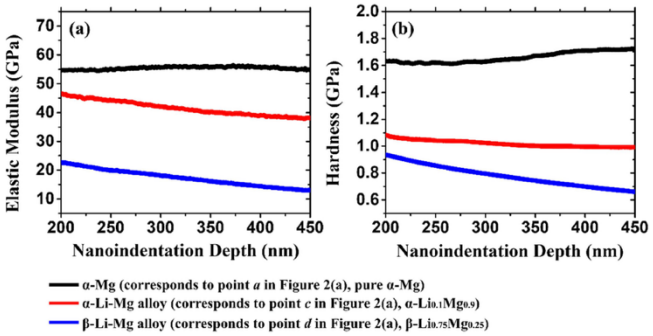


Fig 3. Nanoindentation tests of 3 phases of LiMg alloy: α-Mg, α-LiMg, and β-LiMg. Nanoindentation (a) elastic modulus and (b) hardness vs. depth. [4]

Nanoindentation tests conducted by Shin et al. shown in Fig. 3 indicate that the elastic modulus and hardness of Li-Mg decrease monotonically with increasing Li content: ~55.4 GPa and ~1.66 GPa for α-Mg compared to fully lithiated β-$Li_{0.75}Mg_{0.25}$ with ~17.3 GPa and ~0.78 GPa respectively. Adding Mg to the Li-rich anode reverses this trend, increases bulk hardness from ~2 MPa in pure Li to >57 MPa with 20 at% Mg with the indentation modulus increasing linearly with Mg content (M = 0.44x + 12.2 GPa), shown in Fig. 4 [4, 5].

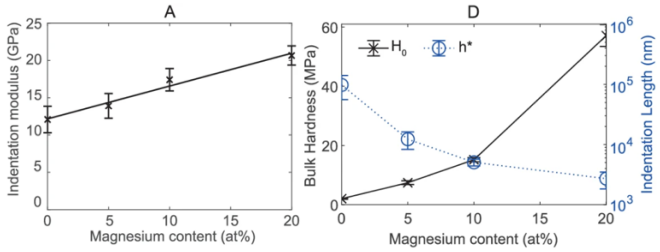


Fig 4. Indentation modulus (A) and bulk hardness (D) increase with Mg content in Li-Mg alloys. [5]

Through the formation of the solid solution, Mg acts as a strengthening agent through solid solution hardening, where Mg atoms distort the Li lattice and resist dislocation motion, driving the increase in hardness observed with increasing Mg content [5].

In the β-phase, LiMg exhibits viscous deformation owing to its low homologous temperature (T/T_m > 0.3 at room temperature). This allows the anode to creep and conform to the surface of the solid electrolyte, reducing voids at the interface which increase resistance. Though, excessive creep at high Mg content reduces stripping capacity [4, 5]. As seen in Fig. 5, creep decreases with increasing Mg content.

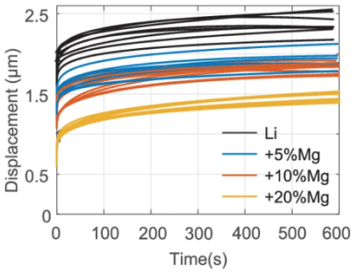


Fig 5. Creep curves showing decreasing viscous deformation with increasing Mg content under constant 1000 μN load [5].

The yield strength of the α- and β-phases, ~0.34 and ~0.26 GPa respectively, are significantly less than the transformation-induced interfacial stresses of ~0.91-2.2 GPa that develop during the full α⇔β transformation, indicating that the anode will plastically deform and could fail every time it crosses the phase boundary. Thus, it is important to cycle a Li-Mg anode only in the β-phase [4].

IV. Phase Behavior and Transformations

The binary Li-Mg phase diagram, shown in Fig. 6, contains two room-temperature phases: the Mg-rich HCP α-phase (0-~15 at% Li) and the Li-Rich BCC β-phase (~30-100 at% Li), with a to phase region where both phases coexist (~10-30 at% Li). This simplistic phase transformation means there is at most one phase transformation per half-cycle, reducing the potential for plastic deformation and mechanical failure [4].

Lithium-Magnesium Alloy

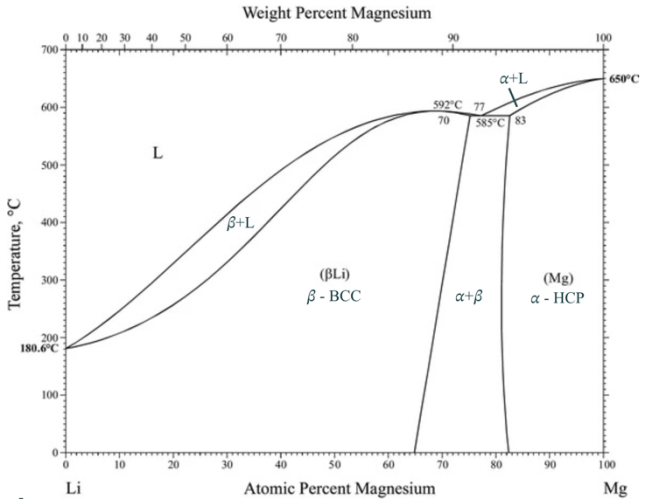


Fig 6. Binary phase diagram for the Li-Mg system (relabeled) [6]

The transformation from the α- to β-phase results in a local volumetric strain of ~16.6% (or a linear strain of ~5.3%) across the phase boundary, generating the aforementioned transformation-induced internal stresses which exceed the yield strength of either phase, thus sufficient to cause plastic deformation, grain refinement, and potential anode cracking during cycling [4]. Volume change during phase transformation is visualized in Fig. 7.

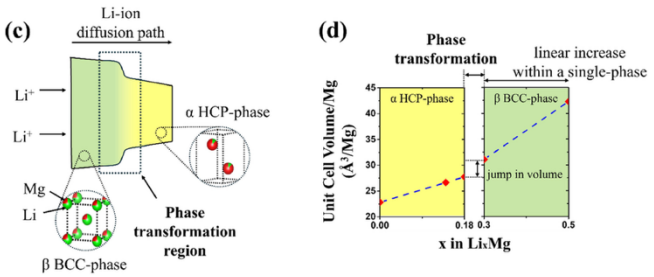


Fig 7. Visualization of volume change with phase transformation (C) and unit volume per Mg atom vs. Li concentration (D) [4].

Cycling in the β-phase, between 0.01 and 0.12 V vs. Li/Li+ is preferable since it avoids the phase boundary thus limiting these phase-induced mechanical stresses. The highly flexible β-BCC crystal structure allows it to hold Li at a wide concentration range (up to 70 at%), enabling cycling. However, limiting cycling to the β-phase does drop capacity from 3,350 mAh/g to ~2,010 mAh/g (experimentally measured 898-1,340 mAh/g), both these levels still far exceed that of graphite anodes used in LIBs (372 mAh/g) [4].

V. Defects and Diffusion Mechanisms

Since vacancies are the main point defect in the Li-Mg substitutional solid solution, diffusion within the BCC Li-Mg lattice is substitutional and vacancy-mediated. Density functional (DFT) calculations predict low kinetically resolved activation (KRA) barriers for nearest-neighbor hops into vacancies for both Li and Mg (~75 meV each), values rivaling the fastest solid-state superionic conductors. However, the equilibrium vacancy concentration at room temperature is extremely low ($N_v/N \approx 10^{-9}$) since the vacancy formation energy is so high (~0.544 eV). Thus, bulk diffusion remains low since the fast kinetics for hopping do not compensate for the scarcity of vacancies [7, 8].

Despite the identical KRA hop barriers in Li and Mg, the diffusion coefficients of both decrease by around one order when they are alloyed together. This was measured by Aspinall et al. in Fig. 8A using Nuclear Magnetic Resonance (NMR), who reported diffusion coefficients of ~9.73×10^{-11} cm^2/s for pure Li compared to ~6.31×10^{-12} cm^2/s for Li + 20 at% Mg at room temperature. This is a result of chemical short-range order (SRO), i.e. Li preferentially neighbors with Mg atoms rather than forming Li-Li or Mg-Mg neighbors. This creates a non-uniform energy landscape in the BCC lattice and raises the effective vacancy-hop energy barrier from ~75 meV to ~197 meV maximum at equiatomic composition [5, 8]. This also results in correlated vacancy motion to avoid unfavorable configuration, which restricts diffusion paths.

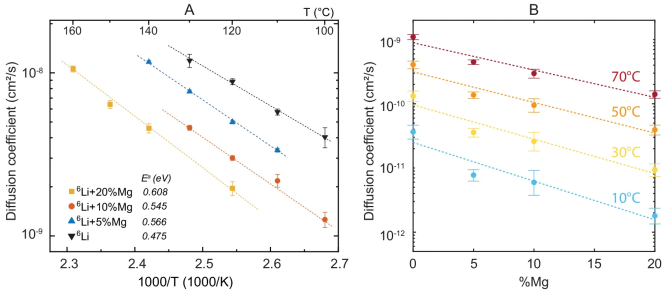


Fig 8. Li diffusion coefficient measured by PFG-NMR decreasing with Mg content (A), and Arrhenius plots showing increasing activation energy with Mg content (B). [5]

This decrease in diffusivity is captured by linearly increasing activation energy with Mg content (E_a = 0.4993 +0.0057x eV), shown in in Fig. 8B, driven by higher vacancy creation enthalpy (due to the alloy's elevated melting point with Mg content ~180→650ºC) and increased hop enthalpy associated with closer lattice packing and greater stiffness. Closer lattice packing with higher Mg levels is supported the linear decline in lattice parameter with increasing Mg content described by a = 3.5095–0.0009x Å (Fig. 9). For anode application, lower Li diffusivity means a slower supply of Li to the interface for stripping, which limits the accessible capacity at a given current density [5]. However, reduced Li diffusion does delay dendrite growth by slowing Li accumulation. So, while more Mg improve interfacial contact retention by increasing hardness and limiting voids via creep, it worsens Li flux to the surface. At low stack pressure (2.5 MPa), a 5 at% Mg alloy outperforms Li since the superior contact retention outweighs the lower diffusivity. But, beyond ~10 at% Mg diffusivity limitation dominates and stripping capacity declines. So, the optimal composition of the Li-Mg anode exists at a balance between these competing effects [5].

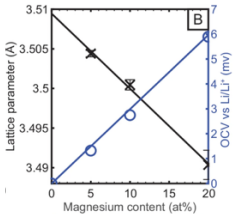


Fig 9. Lattice parameter measured via X-ray diffraction and open circuit voltage (vs. Li) vs at % Mg [5]

VI. Failure Modes

The primary failure mechanism is void formation and coalescence that occurs during Li stripping in SSBs at the anode-electrolyte interface. As Li is extracted faster than it can be replenished by bulk diffusion, vacancies accumulate into voids which grow, merging into a gap which causes contact loss and severe increase in overpotential, i.e. the battery requires more voltage than theoretically predicted due to high resistance at the interface. In pure Li anodes, these leads to a complete delamination at ~1.25 mAh/cm^2, but, in LiMg, voids instead split and collapse rather than coalesce and maintain interfacial contact up to ~3.5 mAh/cm^2. This is the case because Mg atoms preferentially migrate toward the SEI to bond with sulfur from the solid electrolyte, which repels Li vacancies from the surface [10].

Additionally, as mentioned previously, stresses induced by phase transformation result in plastic deformation and grain refinement, evidence by the dramatic drop in x-ray diffraction Bragg peak intensity after a single cycle shown in Fig. 10, indicating significant change in the crystal structure. This deformation accumulates with cycling, eventually degrading the structural integrity of the anode and its capacity retention. However, restricting cycling to the β-phase eliminates this failure mode [4].

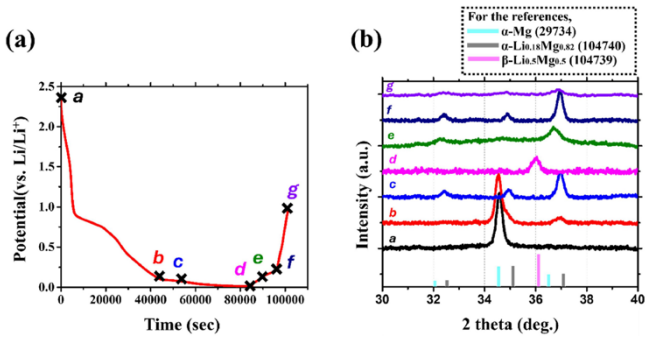


Fig 10. (a) The potential (*vs.* Li/Li$^+$) vs. time; (b) *ex-situ* XRD patterns showing the structural changes in the LiMg anodes during electrochemical cycling with the letters referring to samples taken at the times listed in (a) [4]

Another corrosion-specific failure mode exists in Mg-rich compositions. The natural surface film is porous and fails to protect the metal from salts like chlorine, found in many solid electrolytes. This decay is further accelerated by the high reactivity of lithium, which speeds up the dissolution of the metal. However, when Li content exceeds 15%, a specialized Li-doped protective layer forms and significantly aids corrosion resistance. Understanding this mechanism informs SEI engineering strategies for SSB anodes [10].

VII. Material Origin and Processing

Despite Lithium and Magnesium being globally abundant, their commercial extraction from brines and ores remains an energy-intensive process requiring specialized refining. For laboratory research, these alloys are synthesized using high-temperature melting or via room-temperature mechanical rolling, the latter being a safer, more scalable option for future manufacturing [2, 9]. Alternative methods like electron-beam evaporation or hot-melt-annealing allow for the precise reaction of thin films and single-crystal structures. However, all processing must occur strictly under inert argon atmospheres due to the violent reactivity of Li with moisture [3, 4].

Ultimately, though many fabrication methods exist for Li-Mg alloys, there is a lack of standardized, large-scale manufacturing processes since SSBs and SSB anodes have not been fully commercialized for the mass market. To achieve this commercial transition, manufacturing will have to evolve from small-batch, manual synthesis toward high-output, continuous production lines capable of maintaining extreme atmospheric purity at an industrial scale.

Acknowledgment

The author would like to thank Professor Luisa Chiesa for providing the foundational materials science knowledge necessary to interpret the literature and write this paper and for the valuable feedback provided on earlier components of the project. The author also thanks Ani Gregori Asadourian for her contributions to earlier stages of this project, including research support and reference compilation.

References

[1] A. Joshi, D. K. Mishra, R. Singh, J. Zhang, and Y. Ding, "A comprehensive review of solid-state batteries," *Applied Energy*, vol. 386, p. 125546, May 2025, doi: 10.1016/j.apenergy.2025.125546.

[2] B. Guo *et al.*, "A solid–solution-based Li–Mg alloy for highly stable lithium metal anodes," *Sustainable Energy Fuels*, vol. 6, no. 18, pp. 4137–4145, 2022, doi: 10.1039/D2SE00910B.

[3] L. Zhang *et al.*, "Single-Crystalline LiMg Alloy Anode for Deep Cycling All Solid State Lithium Metal Batteries," *Advanced Functional Materials*, vol. 35, no. 9, p. 2416013, Feb. 2025, doi: 10.1002/adfm.202416013.

[4] J. Shin, Y. Zhang, and M. Pharr, "In-situ mechanical evolution of lithium magnesium alloy anodes in Li-Ion batteries," *Acta Materialia*, vol. 297, p. 121324, Sep. 2025, doi: 10.1016/j.actamat.2025.121324.

[5] J. Aspinall *et al.*, "The impact of magnesium content on lithium-magnesium alloy electrode performance with argyrodite solid electrolyte," *Nature Communications*, vol. 15, no. 1, p. 4511, May 2024, doi: 10.1038/s41467-024-48071-0.

[6] H. Okamoto, "Supplemental Literature Review of Binary Phase Diagrams: Cs-In, Cs-K, Cs-Rb, Eu-In, Ho-Mn, K-Rb, Li-Mg, Mg-Nd, Mg-Zn, Mn-Sm, O-Sb, and Si-Sr," *Journal of Phase Equilibria and Diffusion*, vol. 34, no. 3, pp. 251–263, Jun. 2013, doi: 10.1007/s11669-013-0233-2.

[7] S. S. Behara, J. Thomas, and A. Van der Ven, "Fundamental Thermodynamic, Kinetic, and Mechanical Properties of Lithium and Its Alloys," *Chem. Mater.*, vol. 36, no. 15, pp. 7370–7387, Aug. 2024, doi: 10.1021/acs.chemmater.4c01279.

[8] S. S. Behara and A. Van der Ven, "Role of Short-Range Order on Diffusion Coefficients in the Li–Mg Alloy," *Chem. Mater.*, vol. 36, no. 22, pp. 11236–11245, Nov. 2024, doi: 10.1021/acs.chemmater.4c02334.

[9] L. Zhao *et al.*, "Imaging the evolution of lithium-solid electrolyte interface using operando scanning electron microscopy," *Nature Communications*, vol. 16, no. 1, p. 4283, May 2025, doi: 10.1038/s41467-025-59567-8.

[10] T. W. Cain and J. P. Labukas, "The development of β phase Mg–Li alloys for ultralight corrosion resistant applications," *npj Materials Degradation*, vol. 4, no. 1, p. 17, Jun. 2020, doi: 10.1038/s41529-020-0121-2.